SW Engineering CSC648
StudySync.me
https://github.com/ssohbn/SFSU-csc648-06-Fall26-team04
section 06, team 04
Maeve (team lead)
Henry (frontend lead)
Shreya (scrum master)
Aleks (backend lead)
Jerald (git specialist)
Simran (Ai lead)
Milestone 2
oct 20

1. Data Definitions V2

This should be reasonably consistent with Milestone 1 but should be expanded as needed and refined as per instructors' feedback. Major data items that comprise of sub-data items have to be defined in full (list all its sub-data items, and for images/video list formats, max size etc.). You must use all the data definitions and names consistently in all documents and SW, including, naming for main variables, classes and database elements etc. Focus on data items unique and important to your implementation. Be sure to cover ALL items critical to your project and especially those providing a competitive advantage. At this stage data describing user privileges, and main info (raw data, metadata, supporting data) have to be fully defined (as much as it is possible at this stage)

User:

-username
-bio
-password
-interests
-groupsenrolled

Studygroup:

- Name
- Shortdescription
- Longdescription
- interests
- admins
- Privacy (public, invite/apply, unlisted)
- Members enrolled

2. Functional Requirements V2

Expand functional requirements from Milestone 1 into Milestone 2, with more details as necessary. Keep the same reference numbers with respect to Milestone 1 (i.e. if high level requirement was number R.3. in Milestone 1, then Milestone 2 more detailed requirements of requirement R.3. are R.3.1., R.3.2. etc.)

Prioritize each requirement/spec with 1, 2, 3. (1-must have; 2 – desired; 3 – opportunistic as defined in the class). To develop these priorities on behalf of the user, and making your application complete for usability, marketing and business aspects. The priorities you set later in Milestone 3 will constitute your commitment (especially priorities of 1).

Requirement number	name	priority
1	Sign up	1
1.1	Edit bio	2
1.2	Add interests	1
2	Group system	1
2.1	Create groups	1
2.2	Enroll in groups	1
2.3	Edit groups	2
2.4	Group privacy	2
2.4.1	Open groups	1
2.4.2	Unlisted groups	1
2.4.2	Invite only groups	3
2.4.3	Group application	3
2.5	Group/user Interest enrolling	1
2.5.1	Interest creation	1
3	Canvas calendar integration	2

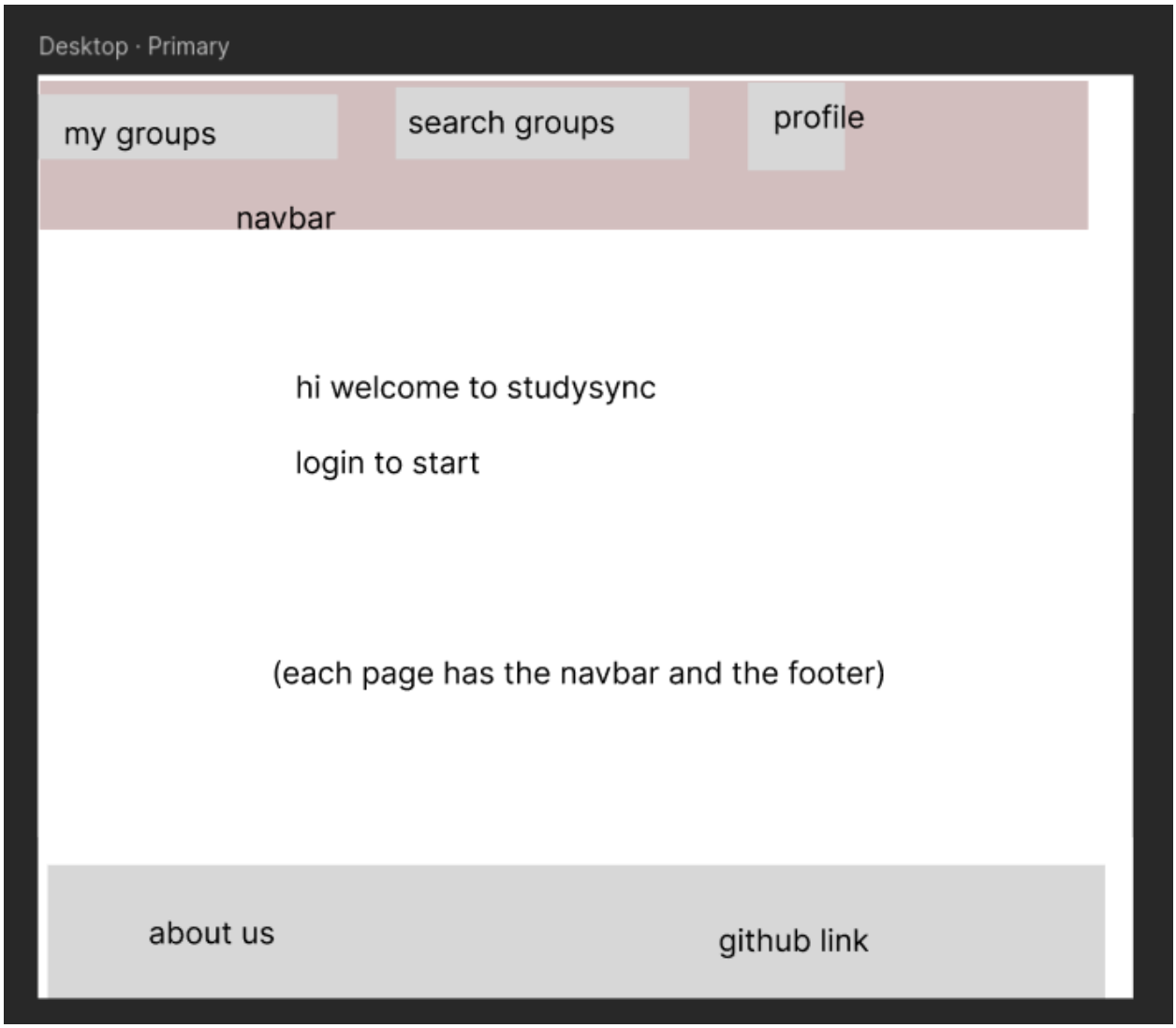
3. UI Mockups and UX Flows

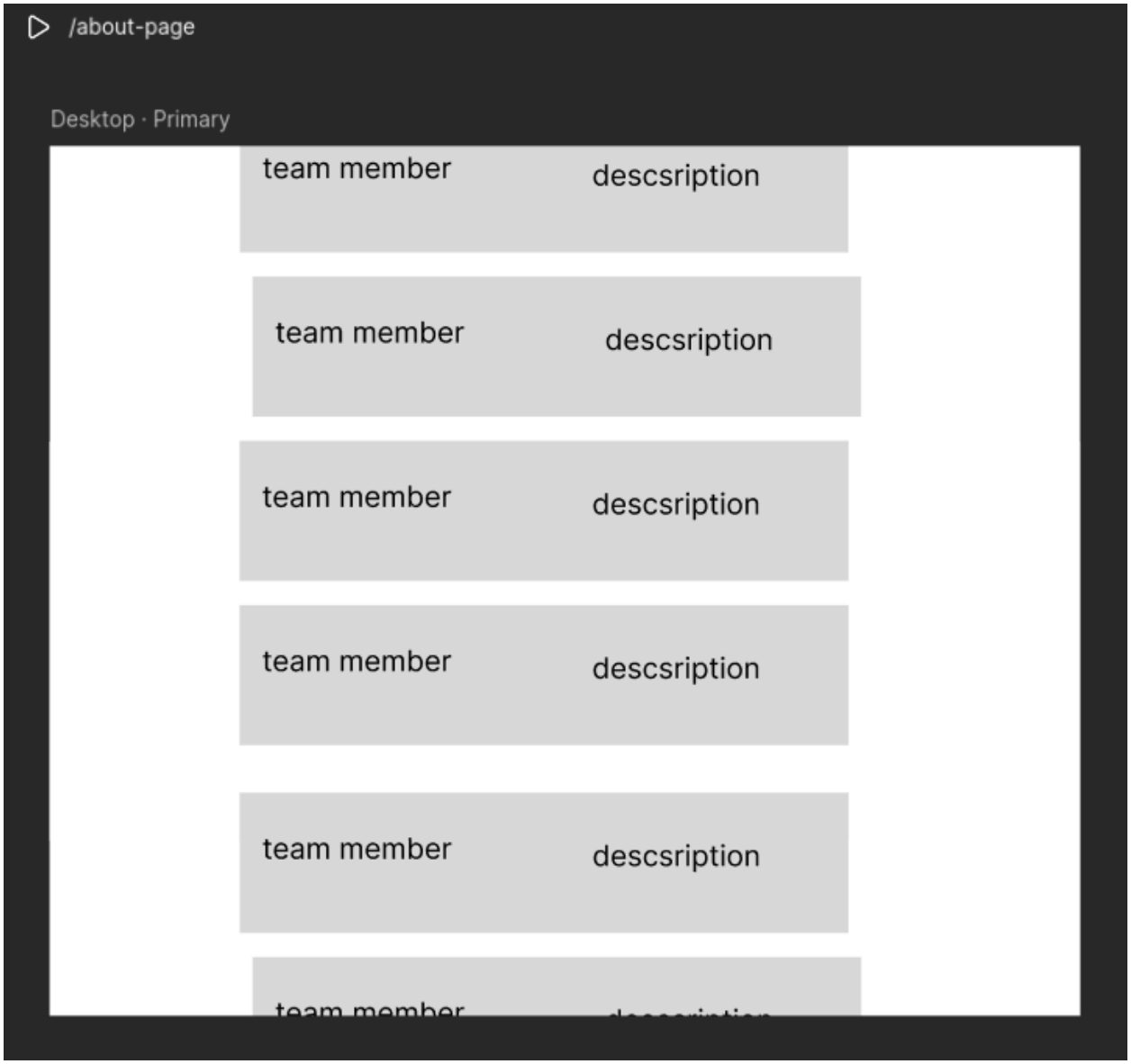
• Create UX prototypes for 5~6 major functional requirements with high priorities, using any drawing tools. Start with black and white wire diagrams focusing on major UX flows. Create a sequence of mockups screenshots for the user stories.

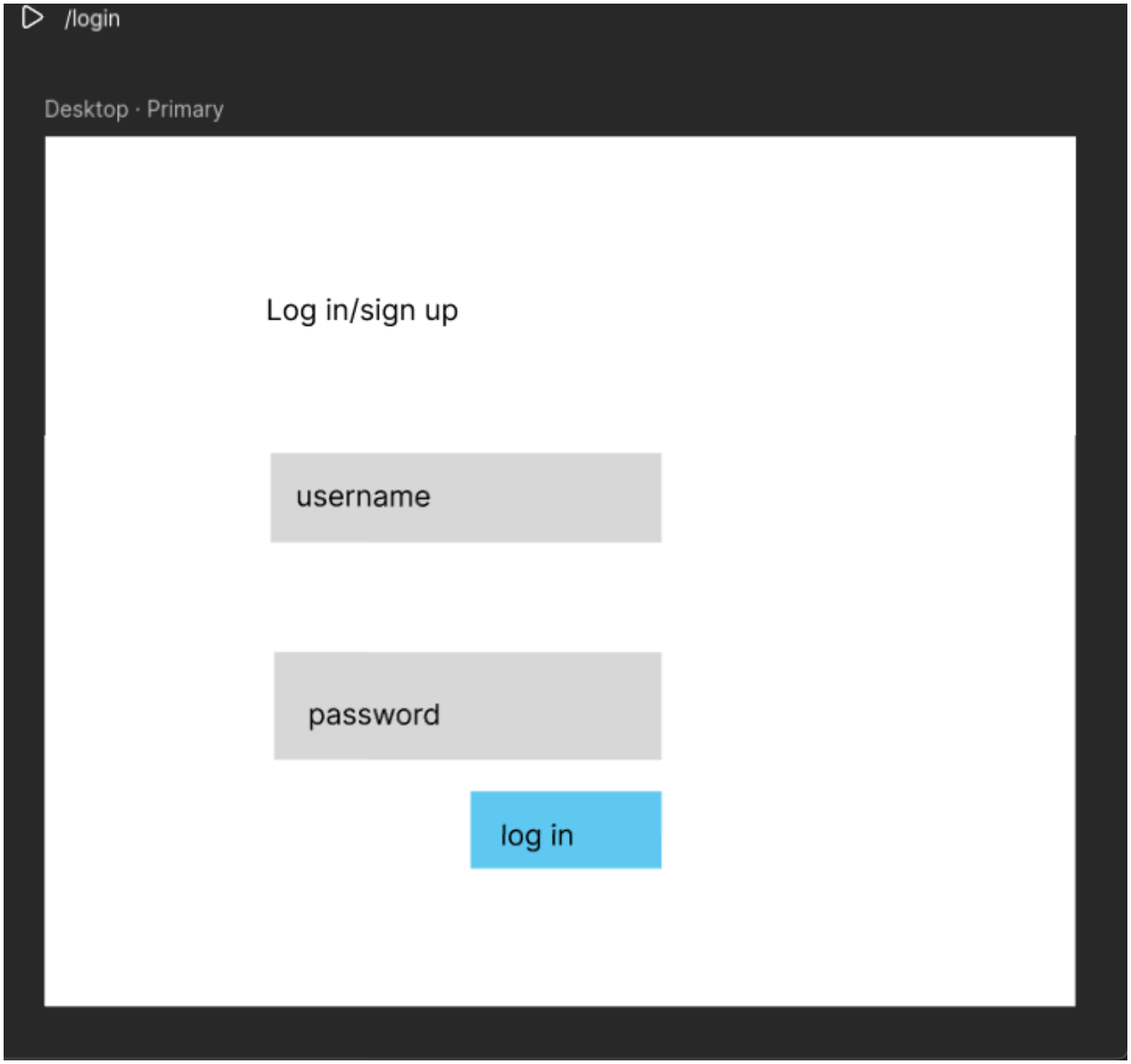
• The format for prototype is flexible. Do not have to use graphics or colors yet (unless absolutely necessary). Instead, it displays basic UI concepts (functions, screen layouts, UX flow…).

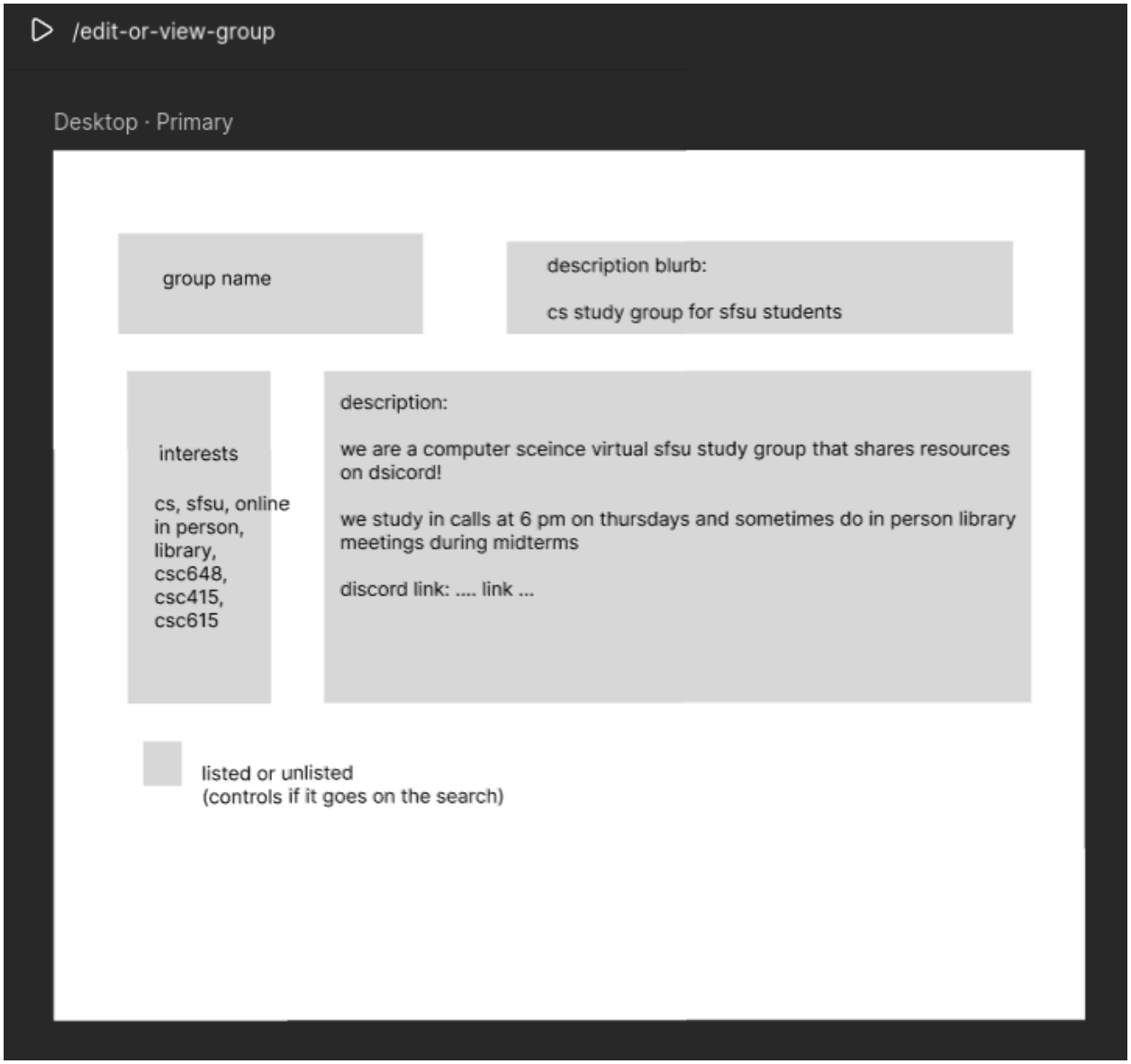
• Once prototype is developed, review to see each of UX principles is met (useful, usable, desirable, accessible, findable, credible) and document it in the M2 documentation.

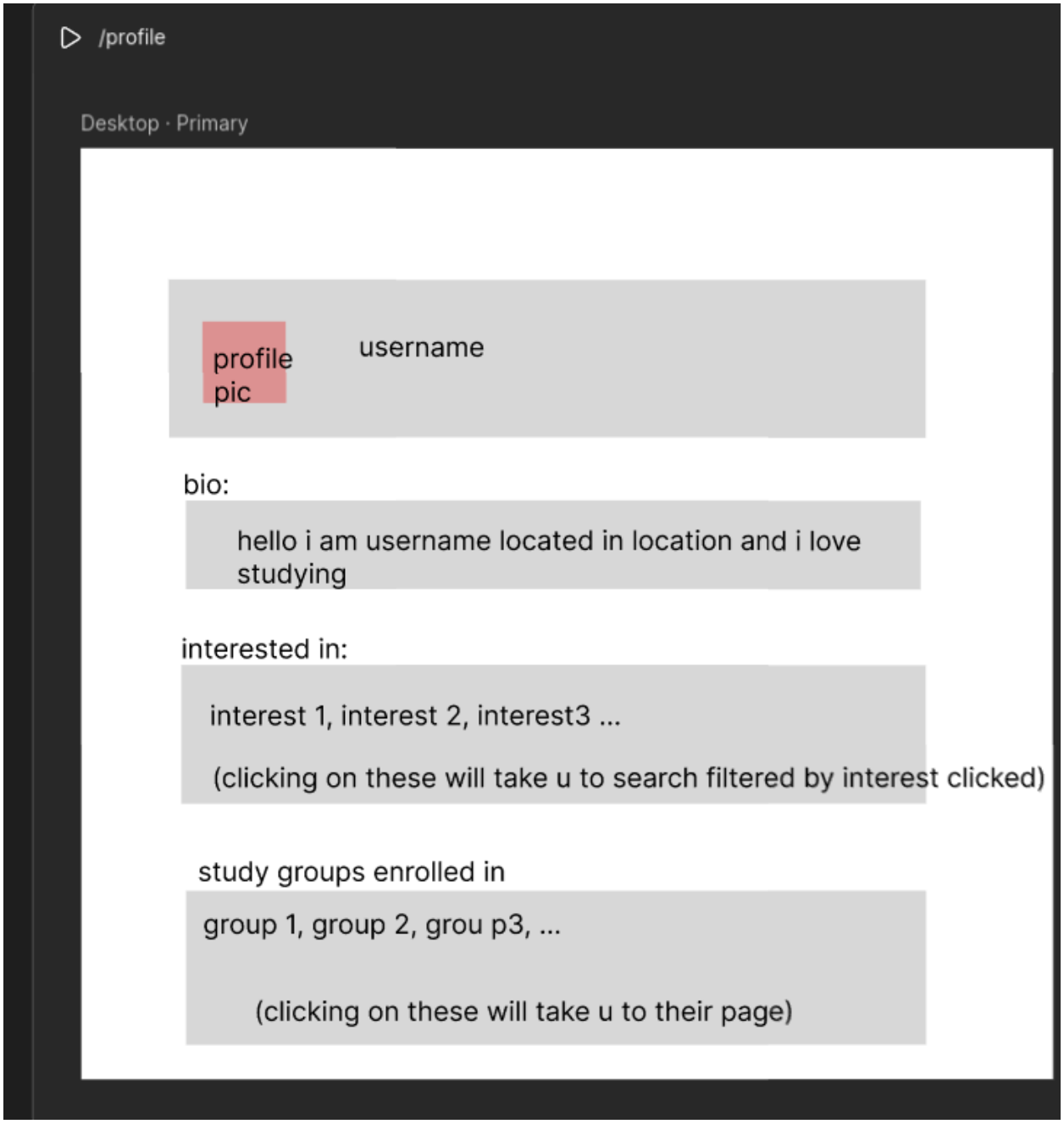
We recommend front-end team be assigned to this task.

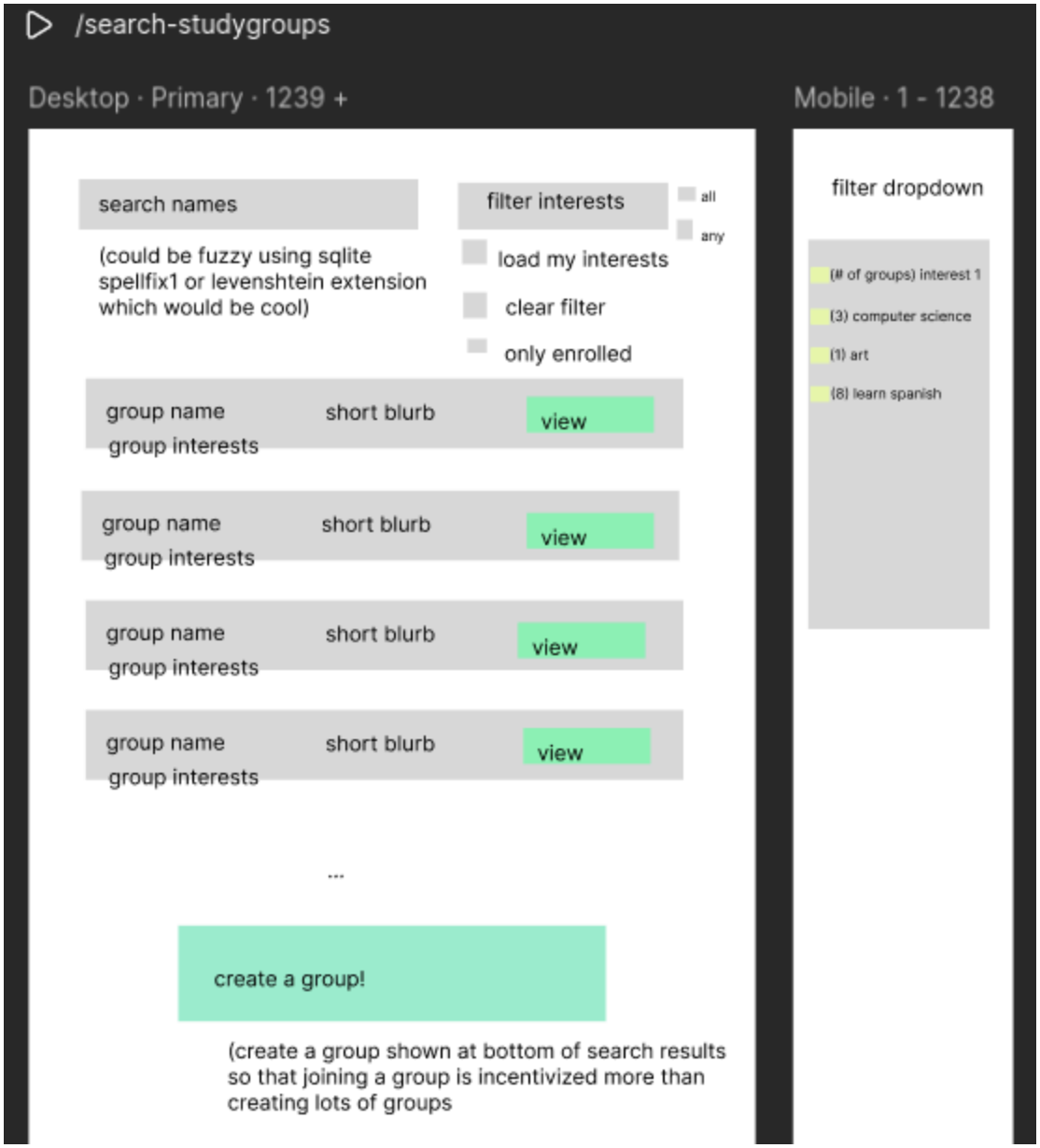


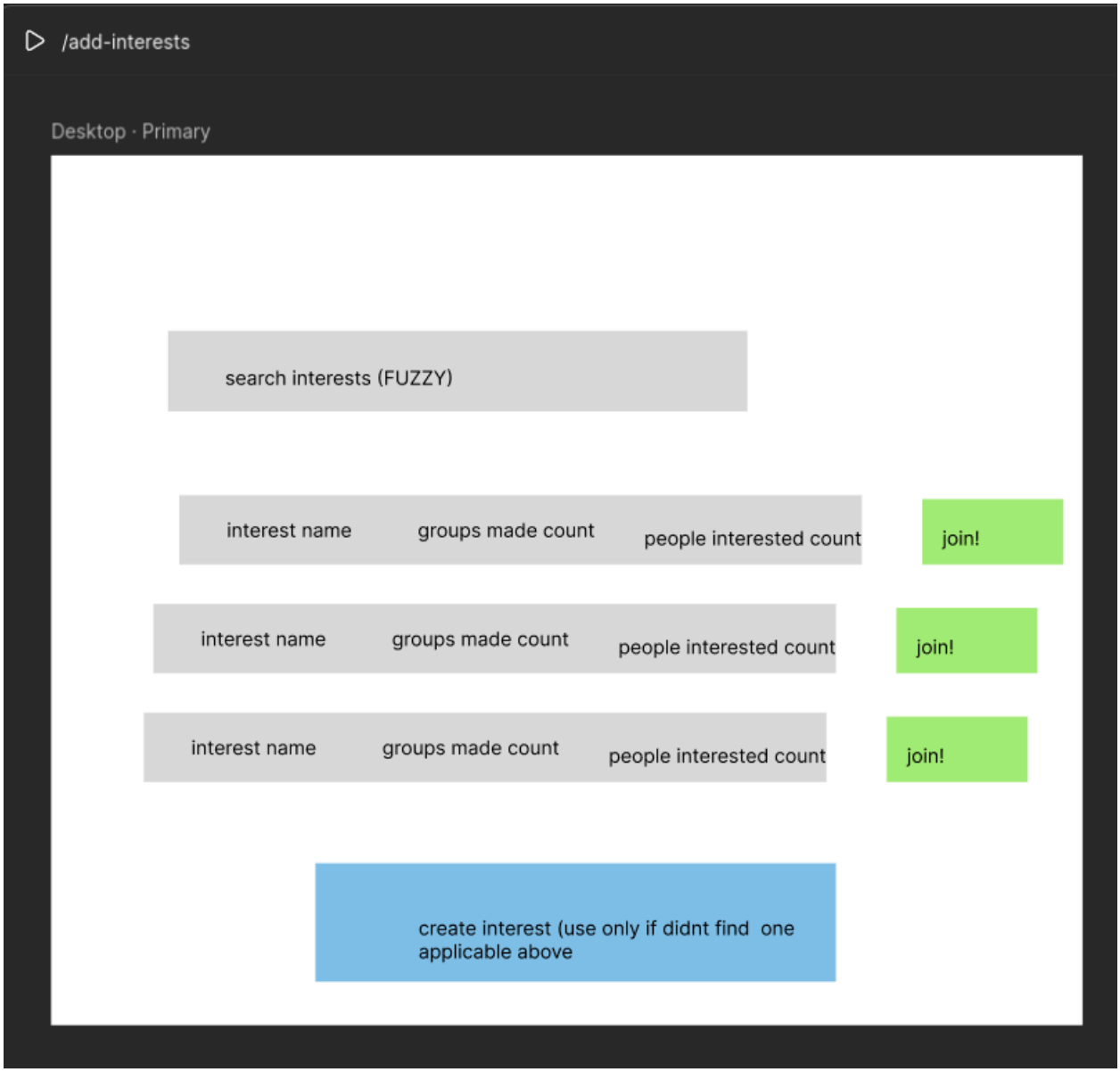

/about-page
Desktop · Primary
team member
descsription
team member
descsription
team member
descsription
team member
descsription
team member
descsription
team member


/login
Desktop · Primary
Log in/sign up
username
password
log in


/edit-or-view-group
Desktop · Primary
group name
description blurb:
cs study group for sfsu students
interests
cs, sfsu, online
in person,
library,
csc648,
csc415,
csc615
description:
we are a computer sceince virtual sfsu study group that shares resources on dsicord!
we study in calls at 6 pm on thursdays and sometimes do in person library meetings during midterms
discord link: .... link ...
listed or unlisted
(controls if it goes on the search)


/profile
Desktop · Primary
profile pic
username
bio:
hello i am username located in location and i love studying
interested in:
interest 1, interest 2, interest3 ...
(clicking on these will take u to search filtered by interest clicked)
study groups enrolled in
group 1, group 2, grou p3, ...
(clicking on these will take u to their page)


/search-studygroups
Desktop · Primary · 1239 +
Mobile · 1 - 1238
search names
(could be fuzzy using sqlite spellfix1 or levenshtein extension which would be cool)
filter interests
all
any
load my interests
clear filter
only enrolled
group name
short blurb
view
group interests
group name
short blurb
view
group interests
group name
short blurb
view
group interests
group name
short blurb
view
group interests
...
create a group!
(create a group shown at bottom of search results so that joining a group is incentivized more than creating lots of groups
filter dropdown
(# of groups) interest 1
(3) computer science
(1) art
(8) learn spanish


/add-interests
Desktop · Primary
search interests (FUZZY)
interest name
groups made count
people interested count
join!
interest name
groups made count
people interested count
join!
interest name
groups made count
people interested count
join!
create interest (use only if didnt find one applicable above

4. High level Architecture, Database Organization
• DB organization: Describe the main database schema/organization (high-level), e.g., list main DB tables (e.g. their titles) and columns in each DB table. For no-SQL database like MongoDB, please define JSON-based documents. Make sure the titles and entry names are easy to understand in plain English and consistent with data definitions in Section 1 above.
• Add/Delete/Search architecture:
• For the functional requirements, specify the operations permitted for each SQL table or MongoDB collection (e.g., add, search, update, delete, display) and the data fields involved (e.g., name, email, user ID).
• For example: Users table — add a user (name, email), search for a user by email, update a user's name, delete a user (by email), and display user information (name, email).
• APIs
• Backend endpoint APIs : describe and define major backend APIs that you will create to communicate between the frontend and backend.
• If you are using the 3rd party API such as three.js or Niantic Spatial SDK, please describe them in your architecture.
• If you are using open-source components, please describe them in your architecture.
• If you have changed SW tools and frameworks or added any new one, please describe it.

We have added htmx to our frontend to bridge the backend and frontend. Htmx lets us write every javascript interaction that most websites use via tags like html. It works well for simple statically rendered sites like ours, and helps us make it interactive without writing lots of javascript or complicating our tech stack.

StudyGroups
-id
-Name
-shortdescription
-longdescription

User
-id
-name
-bio
-password

Enrollment
-Userid
-Studygroupid
-Member Role (admin/regular)

Interest
-id
-Name

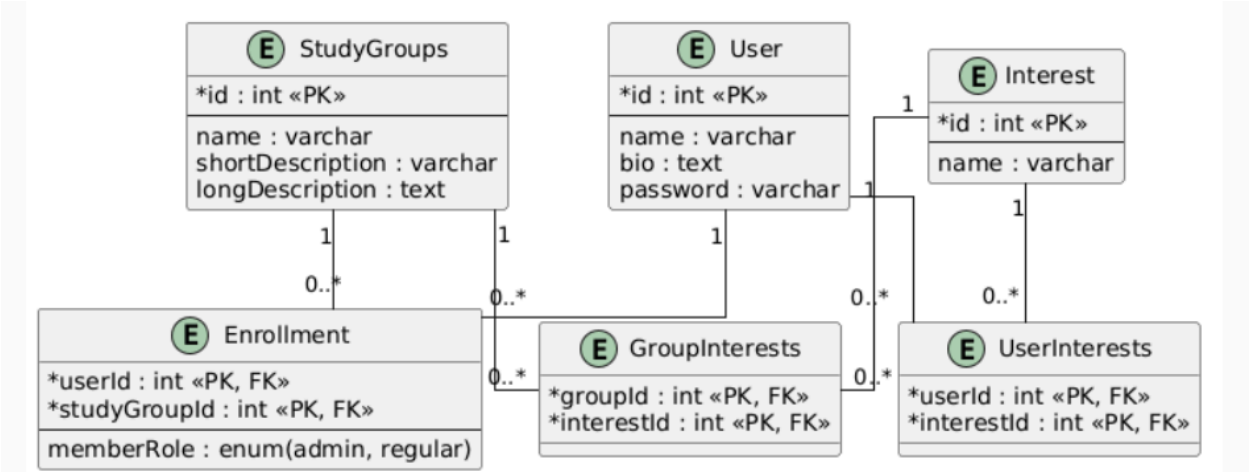
userInterests
-userid
-interestid

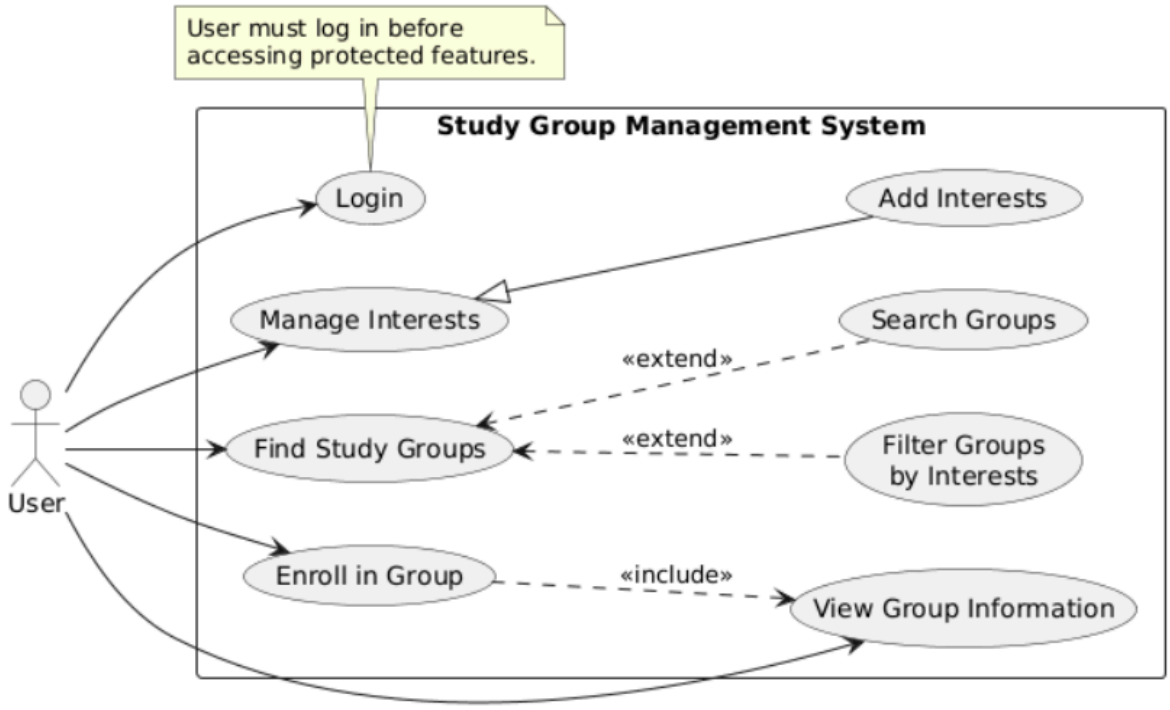
Groupinterests
-groupid
-interestid

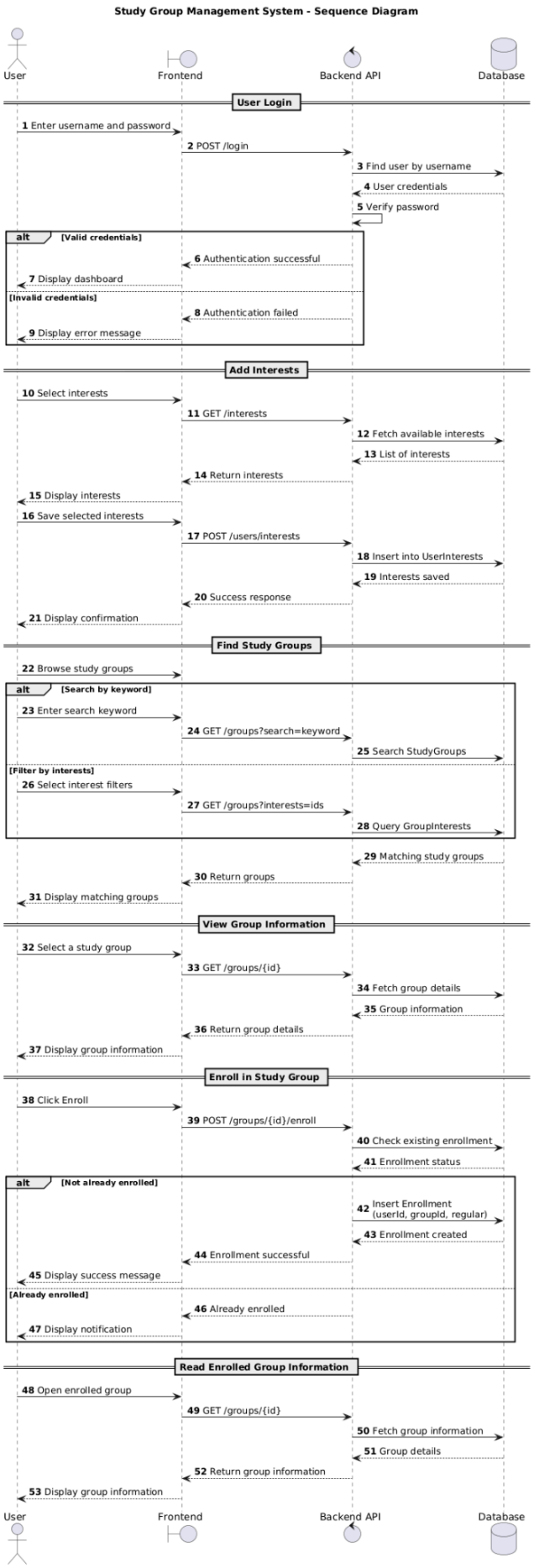
We will have apis to create read update and delete studygroups
We will have apis to create read update delete user
We will have apis to create read update and delete enrollment
We will have apis to create and read interests (it doesnt matter if theyre permanent)
We will have apis to create read update and delete userinterests
We will have apis to create read update and delete groupinterests

5. High Level UML Diagrams
Familiarize yourself with Unified Modeling Language (UML). For Milestone 2, provide:
High-level UML class diagrams for implementation classes of core functionality, Focus on main high-level classes only (one or at most two levels deep). This must reflect an OO approach to implementing your web site.
High-level sequence diagrams: for ~5~6 functional requirements, please develop UML sequence diagram.

Uml diagrams made with ai generated plantuml






Study Group Management System - Sequence Diagram
User
Frontend
Backend API
Database
User Login
1 Enter username and password
2 POST /login
3 Find user by username
4 User credentials
5 Verify password
alt [Valid credentials]
6 Authentication successful
7 Display dashboard
[Invalid credentials]
8 Authentication failed
9 Display error message
Add Interests
10 Select interests
11 GET /interests
12 Fetch available interests
13 List of interests
14 Return interests
15 Display interests
16 Save selected interests
17 POST /users/interests
18 Insert into UserInterests
19 Interests saved
20 Success response
21 Display confirmation
Find Study Groups
22 Browse study groups
alt [Search by keyword]
23 Enter search keyword
24 GET /groups?search=keyword
25 Search StudyGroups
[Filter by interests]
26 Select interest filters
27 GET /groups?interests=ids
28 Query GroupInterests
29 Matching study groups
30 Return groups
31 Display matching groups
View Group Information
32 Select a study group
33 GET /groups/{id}
34 Fetch group details
35 Group information
36 Return group details
37 Display group information
Enroll in Study Group
38 Click Enroll
39 POST /groups/{id}/enroll
40 Check existing enrollment
41 Enrollment status
alt [Not already enrolled]
42 Insert Enrollment (userId, groupId, regular)
43 Enrollment created
44 Enrollment successful
45 Display success message
[Already enrolled]
46 Already enrolled
47 Display notification
Read Enrolled Group Information
48 Open enrolled group
49 GET /groups/{id}
50 Fetch group information
51 Group details
52 Return group information
53 Display group information
User
Frontend
Backend API
Database

6 .Identify actual key risks for your project at this time Identify only actual and specific risks in your current work such as

• skills risks and mitigation plan
o Do you have a proper study plan to cover all the necessary technologies?

• schedule risks
o Does your team have a team schedule for every member including their detailed task?
o If change happens, does it update transparently? Does your team use project management tool (e.g. Jira, Trello)

• teamwork risks (any issues related to teamwork);
o Everybody is on the meeting regularly?
o Everybody keeps his/her pace? If not, what is your plan to mitigate the risks?

• legal/content risks (can you obtain content/SW you need legally with proper licensing, copyright).

Tell us how do you plan to resolve each actual risk you have. The key is to resolve risks as soon as possible. Categorizing risk as above helps a lot in managing them. Be brief: identify the risk and explain (2-3 lines), list how you will address this issues' (2-3 lines), and share with every team members.

We shouldn't have many risks going forward, a few people got sick this week and alongside midterms we have done less for this milestone, but as we get better and get less busy we can spend more time on the milestones. Everyone has been on pace for the milestones and there is no content we need licensing for. Our project is simple so that studying needed is minimal, and the study plan from the 0th milestone is still valid.

We do not have a jira or trello but we have meetings and communication in meetings are posted in the meetings doc and on discord

7. Project management
Describe in no more than half a page how your team managed M2 tasks including
• how each member's progress is being shared in scrum meeting? Is it transparently shared?
• Do you have a tool to manage each member's task, such as Notion, Jira? Strongly recommend such tools

We work together during the meetings and discuss what we are doing as we do it. Progress is shared by seeing the work and showing it to each other. We don't have any tools to manage tasks. We notify each other when tasks are complete or being worked on via a group chat, or by organizing ourselves in our meetings.